

Avoiding duplication, the given expression is equivalent to  $(p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge r)$

If we introduce ordering among variables, as  $p_1, p_2, \dots, p_n$  then there are  $2^n$  minterms and they can be assigned values from 0 to  $2^n - 1$ . The number  $i$  corresponds to the minterm  $m_i$  as follows.

Let  $b_1, b_2, \dots, b_n$  be the binary representation of  $i$ . Then in the minterm  $m_i$ , the variable  $p_j$  occurs as  $p_j$  if  $b_j = 1$  and occurs as  $\neg p_j$  if  $b_j = 0$ .

00...0 corresponds to  $\neg p_1 \wedge \neg p_2 \wedge \dots \wedge \neg p_n$   
11...1 corresponds to  $p_1 \wedge p_2 \wedge \dots \wedge p_n$

Example 3:  $\Sigma(0, 2, 3)$

Example 4:  $\Sigma(1, 3, 6, 7)$

### 1.9.3 Principal Conjunctive Normal Form

- We define maxterm as a dual to minterm.
- For a given number of variables, the maxterm consists of disjunctions in which each variable or its negation, but not both, appears only once.
- It can be seen that each of the maxterms has the truth value F for exactly one combination of the truth values of the variables.  $\nabla \text{Imp}$

| $p$ | $q$ | $p \vee q$ | $\neg p \vee \neg q$ | $p \vee \neg q$ | $\neg p \vee q$ |
|-----|-----|------------|----------------------|-----------------|-----------------|
| T   | T   | T          | F                    | T               | F               |
| T   | F   | T          | T                    | F               | T               |
| F   | T   | T          | T                    | T               | F               |
| F   | F   | F          | F                    | T               | T               |

**Definition 4:** For a given formula, an equivalent formula consisting of conjunction of maxterms only is known as its principal conjunctive normal form. This normal form is also called the product of sums canonical form.

**Example 5:** Find principal conjunctive normal form for  $(p \rightarrow q)$

$$(p \rightarrow q) \equiv (p \rightarrow q) \wedge (q \rightarrow p) \quad (\text{Section 1.3, Table 8 Equivalence 1})$$

$$\equiv (\neg p \vee q) \wedge (\neg q \vee p) \quad (\text{Conditional-disjunction equivalence})$$